DATA ANALYTICS PORTFOLIO PROJECT

By

AMY MUNJOMA

# Global Sales & Profitability Analysis

*An interactive financial performance dashboard built with Microsoft Power BI*

Tools: Microsoft Power BI • DAX • Data Visualization

## PROJECT SNAPSHOT

| | | |
|---|---|---|
| **$118.73M** Total Sales | **$16.89M** Total Profit | **14.23%** Profit Margin |
| **$9.21M** Total Discounts | **1.13M** Units Sold | **$101.83M** Total COGS |

This case study documents the business questions, DAX measures, dashboard structure, key findings, and recommended follow up analyses developed for the project.

# 1. Project Overview

This project uses Microsoft Power BI to explore financial performance across products, countries, customer segments, discount bands, and time periods. The goal is to turn transaction level financial data into an interactive report that helps users examine sales, costs, discounts, and profitability together.

### Project objectives

- Monitor headline financial and operational KPIs.
- Compare profitability across countries and products.
- Explore sales contribution and profit margins by customer segment.
- Review monthly and quarterly performance patterns.
- Identify areas that warrant further investigation and evidence based business recommendations.

### Tools and capabilities

Microsoft Power BI for dashboard development and interactive reporting; DAX for reusable measures and financial calculations; and data visualization techniques to communicate performance patterns.

# 2. Business Problem

Organizations need more than revenue totals to understand business performance. Sales may be strong while profitability varies by segment, product, or market. Discounting may support sales volume, but its contribution to profit needs to be assessed rather than assumed.

The dashboard addresses the following questions:

- How much sales and profit does the business generate, and what is its overall profit margin?
- How do profit and discounts vary by country and product?
- Which customer segments contribute most to sales, and which show stronger margins?
- How does performance change by month, quarter, and year?

**Central business question:** How do sales, costs, discounts, products, markets, and customer segments interact and where should further analysis focus to improve financial performance?

# 3. Dataset & Analytical Dimensions

The dashboard uses the Microsoft Financials sample data, as identified during project development. The visuals show five countries, six products, five customer segments, discount bands, and time based fields. Confirm the exact dataset version and field definitions before public release.

### Dimensions visible in the dashboard

- Geography: Canada, France, Germany, Mexico, and United States.
- Products: Paseo, VTT, Velo, Amarilla, Montana, and Carretera.
- Customer segments: Government, Small Business, Enterprise, Midmarket, and Channel Partners.
- Time: Year, month, and quarter.
- Discount bands: High, Medium, Low, and None.

# 4. Data Preparation & DAX Calculations

DAX measures were created to provide reusable financial and operational calculations for the report. These measures support KPI cards and comparisons across the dashboard's filter context.

## Measures created

| Measure | Analytical purpose |
|---|---|
| Average Manufacturing Price | Average manufacturing price in the selected context. |
| Average Sale Price | Average selling price in the selected context. |
| Discounts (%) | Percentage-based discount metric; exact denominator should be documented from the DAX formula. |
| Profit Margin (%) | Profit relative to sales, expressed as a percentage. |
| Total COGS | Aggregated cost of goods sold. |
| Total Discounts | Aggregated discount value. |
| Total Gross Sales | Gross sales before the relevant deductions, according to the measure definition. |
| Total Profit | Aggregated profit. |
| Total Sales | Aggregated sales. |
| Total Units Sold | Aggregated quantity sold. |

The report uses slicers to support interactive exploration by year, product, country, quarter, and discount band where available. Before publishing, verify the model structure, any Power Query steps, and the exact formulas rather than claiming unconfirmed transformations.

# 5. Dashboard & Analysis

## Page 1 — Overall Performance

The first page summarizes headline financial and operational KPIs and provides context through quarterly profit, sales by segment, units sold by product, and transaction distribution by discount band.

- Headline KPIs: $118.73M sales, $101.83M COGS, $16.89M profit, 14.23% profit margin, $9.21M discounts, and 1.13M units sold.
- Quarterly profit: Q4 2014 is the strongest quarter shown, at approximately $4.4M.
- Sales by segment: Government contributes approximately $53M and Small Business approximately $42M.
- Units by product: Paseo leads at approximately 338K units.

## Page 2 — Product & Geographic Performance

The second page compares profit by country, product contributions within markets, discounts by product, and discounts by country. It helps users examine where profit is generated and how discount exposure differs across markets and products.

- Profit by country: France leads at approximately $3.8M, followed by Germany at $3.7M.
- Discounts by product: Paseo has the highest total discount value, approximately $2.60M.
- Discounts by country: The United States shows approximately $2.24M in discounts.

## Page 3 — Segment Performance

The third page examines monthly sales, quarterly profit by segment, and profit margin by segment. This allows revenue patterns to be compared with profitability.

- Monthly sales: October is the strongest month shown, at approximately $22M.

- Segment margins: Channel Partners show a 73.13% margin; Enterprise shows -3.13%.

## 6. Key Findings

- Overall performance: The dashboard reports $118.73M in sales, $16.89M in profit, and a 14.23% profit margin.
- Sales concentration: Government is the largest sales segment, at approximately $53M, followed by Small Business at approximately $42M.
- Margin differences: Channel Partners have the highest displayed margin (73.13%), while Enterprise has a negative displayed margin (-3.13%).
- Geographic variation: France has the highest displayed total profit among the five countries, at approximately $3.8M.
- Discount exposure: Paseo leads both units sold and total discount value, suggesting that discount effectiveness is worth investigating.
- Seasonality or timing pattern: October has the highest monthly sales value shown. The cause is not established by the chart alone.

**Core analytical takeaway:** The factors driving revenue are not necessarily the same factors driving profitability. Sales, costs, discounts, and margins should be evaluated together.

## 7. Business Recommendations

1. Investigate Enterprise profitability. Review product mix, pricing, COGS, discounts, and customer level results to identify the drivers of the negative displayed margin.
2. Evaluate discount effectiveness. Compare discount levels with sales volume and profit to determine whether discounting is generating sufficient commercial value.
3. Study Channel Partners' margin. Validate the calculation and investigate the mix, costs, prices, and discounts associated with the high displayed margin before attempting to replicate the result.
4. Compare geographic performance. Examine product and segment mix, pricing, costs, and discounts in higher and lower profit markets.
5. Review Paseo's discount strategy. Assess whether the product's discount exposure is justified by its sales and profit contribution.
6. Investigate the October sales peak. Check the contributing products, countries, segments, and discount bands before attributing the increase to any particular cause.

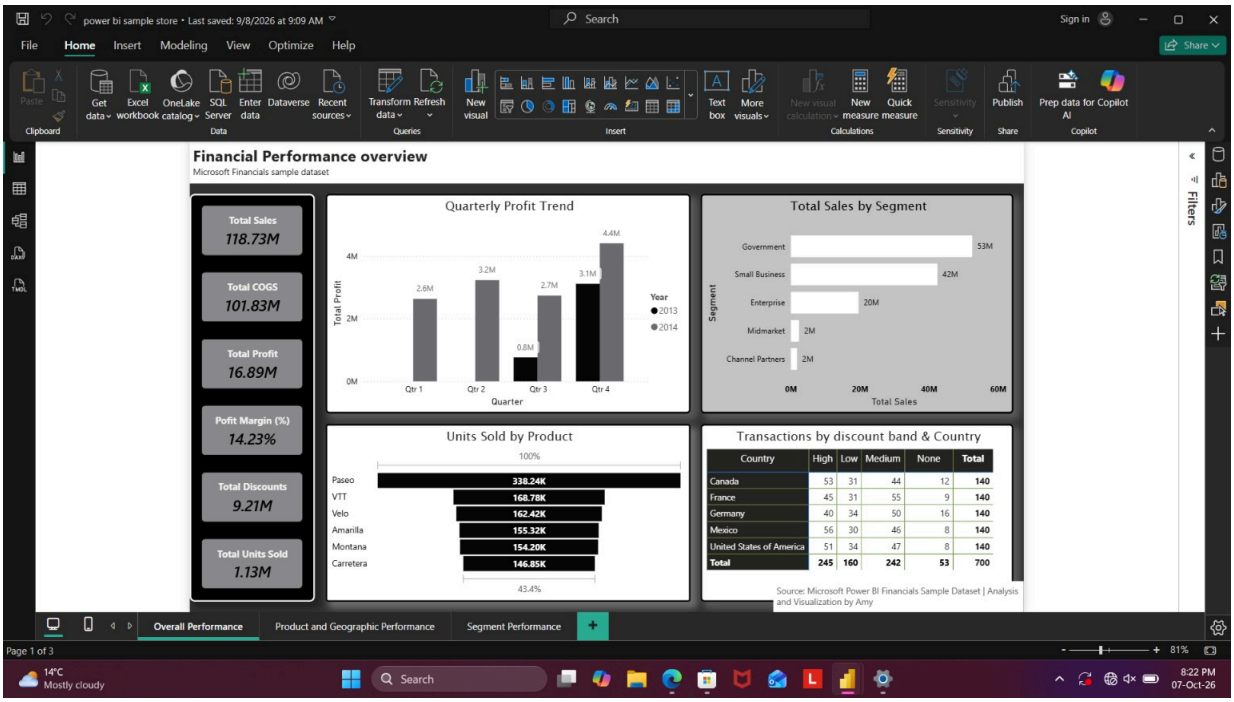
## 8. Skills Demonstrated

Technical: Microsoft Power BI; DAX measures; KPI development; interactive slicers; multi-page dashboard design; financial calculations; data visualization.

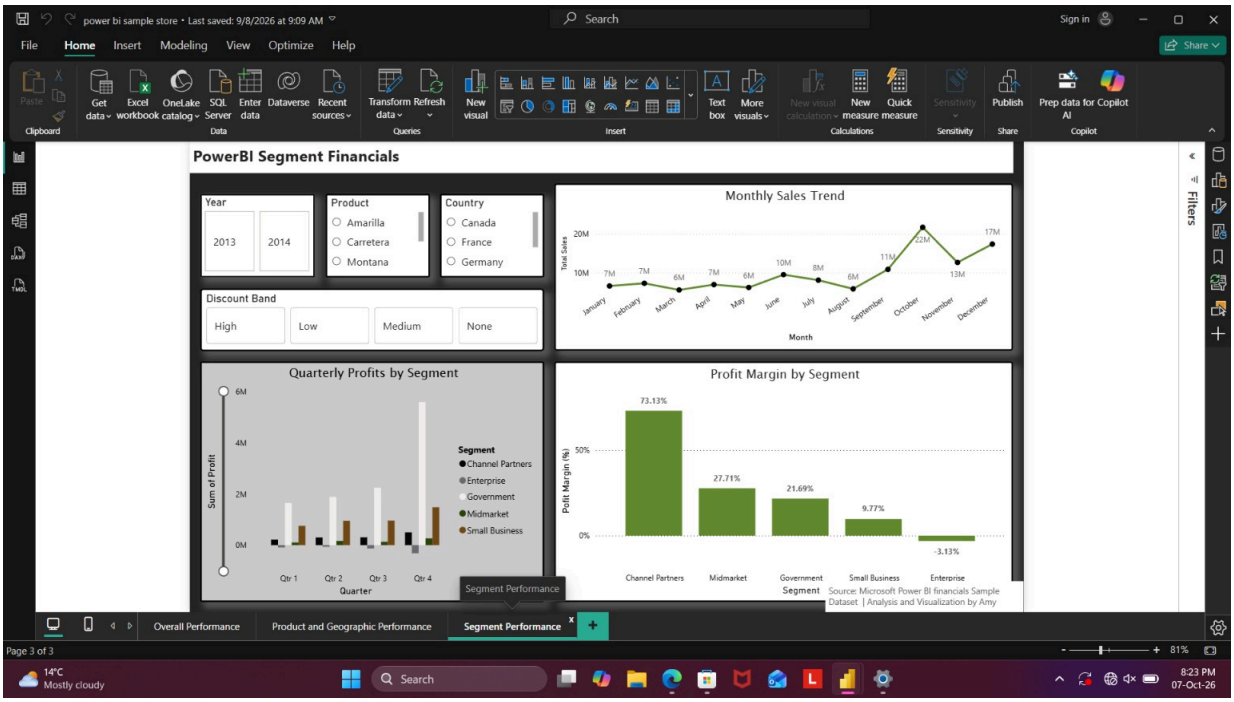
Analytical: Sales and profitability analysis; product and geographic comparisons; segment analysis; monthly and quarterly trend analysis; discount analysis; insight generation.

Business: Framing analytical questions; identifying results that warrant further investigation; communicating findings; translating observations into practical recommendations.

# Appendix: Dashboard Screenshot 1



# Appendix: Dashboard Screenshot 2



# Appendix: Dashboard Screenshot 3

